6.3 Group-valued flows

Let $G = (V, E)$ be a multigraph. If f and g are two circulations on G with values in the same abelian group H, then $(f+g)\colon \vec{e} \mapsto f(\vec{e}) + g(\vec{e})$ and $-f\colon \vec{e} \mapsto -f(\vec{e})$ are again circulations. The circulations on G with values in H thus form a group in a natural way. $f+g$ $-f$

An *H-flow* in our terminology[5] is a circulation $f\colon \vec{E} \to H$ that is *nowhere zero*, one that satisfies $f(\vec{e}) \neq 0$ for all $\vec{e} \in \vec{E}$. Note that the set of H-flows on G is not closed under addition: if two H-flows add up to zero on some oriented edge $\vec{e}$, then their sum is no longer an H-flow. By Corollary 6.1.2, a graph with an H-flow cannot have a bridge. *H-flow*

For finite groups H, the number of H-flows on G – and, in particular, their existence – surprisingly depends only on the order of H, not on H itself:

Theorem 6.3.1. (Tutte 1954)
For every multigraph G there exists a polynomial P such that, for any finite abelian group H, the number of H-flows on G is $P\big(|H|-1\big)$.

(6.1.1) (6.1.3)

Proof. Let $G =: (V, E)$; we use induction on $m := |E|$. Let us assume first that all the edges of G are loops. Then, given any finite abelian group H, every map $\vec{E} \to H \setminus \{0\}$ is an H-flow on G. Since $|\vec{E}| = |E|$ when all edges are loops, there are $\big(|H|-1\big)^m$ such maps, and $P := x^m$ is the polynomial sought.

Now assume there is an edge $e_0 = xy \in E$ that is not a loop; let $\vec{e_0} := (e_0, x, y)$ and $E' := E \setminus \{e_0\}$. We consider the multigraphs $e_0 = xy$ E'

$$G_1 := G - e_0 \quad \text{and} \quad G_2 := G/e_0\,.$$

By the induction hypothesis, there are polynomials P_i for $i = 1, 2$ such that, for any finite abelian group H and $k := |H| - 1$, the number of H-flows on G_i is $P_i(k)$. We shall prove that the number of H-flows on G equals $P_2(k) - P_1(k)$; then $P := P_2 - P_1$ is the desired polynomial. P_1, P_2 k

Let H be given, and let F denote the set of all H-flows on G. Our aim is to show that H, F

$$|F| = P_2(k) - P_1(k)\,. \tag{1}$$

The H-flows on G_1 are precisely the restrictions to $\vec{E'}$ of those H-circulations on G that are zero on e_0 but nowhere else. Let us denote the set of these circulations on G by F_1; then F_1

$$|F_1| = P_1(k).$$

[5] To avoid cumbersome repetitions of the phrase 'nowhere zero' before 'H-flow', we deviate slightly here from standard terminology. See the footnote in the notes.

Let F_2 denote the set of H-circulations on G that are non-zero except *possibly* on e_0. The H-flows f on G_2 define circulations $f' \in F_2$ on G as in the proof of Lemma 6.1.3, where $\varphi\colon f \mapsto f'$ is injective. This map φ is also surjective: use Proposition 6.1.1 with $X = \{x, y\}$ to check (F2) at v_{e_0} for the map f induced naturally on $\vec{E}(G/e_0)$ by a given $f' \in F_2$, and note that φ sends f to f' since f' is determined uniquely on $\vec{e_0}$ and $\overleftarrow{e_0}$ by its values on $\vec{E} \setminus \{\vec{e_0}, \overleftarrow{e_0}\}$. Thus,

$$|F_2| = P_2(k).$$

As $F_1 \subseteq F_2$ and $F = F_2 \setminus F_1$, we have (1) as desired. $\square$

flow polynomial

The polynomial P of Theorem 6.3.1 is known as the *flow polynomial* of G.

[6.4.5]
[6.6.1]

Corollary 6.3.2. *If H and H' are two finite abelian groups of equal order, then G has an H-flow if and only if G has an H'-flow.* $\square$

Corollary 6.3.2 has fundamental implications for the theory of algebraic flows: it indicates that crucial difficulties in existence proofs of H-flows are unlikely to be of a group-theoretic nature. On the other hand, being able to choose a convenient group can be quite helpful; we shall see a pretty example for this in Proposition 6.4.5.

k
k-flow

Let $k \geqslant 1$ be an integer and $G = (V, E)$ a multigraph. A $\mathbb{Z}$-flow f on G such that $0 < |f(\vec{e})| < k$ for all $\vec{e} \in \vec{E}$ is called a *k-flow*. Clearly, any k-flow is also an ℓ-flow for all $\ell > k$. Thus, we may ask which is the least integer k such that G admits a k-flow – assuming that such a k exists. We call this least k the *flow number* of G and denote it by $\varphi(G)$; if G has no k-flow for any k, we put $\varphi(G) := \infty$.

flow number
$\varphi(G)$

The task of determining flow numbers quickly leads to some of the deepest open problems in graph theory. We shall consider these later in the chapter. First, however, let us see how k-flows are related to the more general concept of H-flows.

σ_k

There is an intimate connection between k-flows and $\mathbb{Z}_k$-flows. Let σ_k denote the natural homomorphism $i \mapsto \bar{i}$ from $\mathbb{Z}$ to $\mathbb{Z}_k$. By composition with σ_k, every k-flow defines a $\mathbb{Z}_k$-flow. As the following theorem shows, the converse holds too: from every $\mathbb{Z}_k$-flow on G we can construct a k-flow on G. In view of Corollary 6.3.2, this means that the general question about the existence of H-flows for arbitrary groups H reduces to the corresponding question for k-flows.

[6.4.1]
[6.4.2]
[6.4.3]
[6.4.5]

Theorem 6.3.3. (Tutte 1950)
A multigraph admits a k-flow if and only if it admits a $\mathbb{Z}_k$-flow.

Proof. Let g be a $\mathbb{Z}_k$-flow on a multigraph $G = (V, E)$; we construct a k-flow f on G. We may assume without loss of generality that G has no loops. Let F be the set of all functions $f\colon \vec{E} \to \mathbb{Z}$ that satisfy (F1), $|f(\vec{e})| < k$ for all $\vec{e} \in \vec{E}$, and $\sigma_k \circ f = g$; note that, like g, any $f \in F$ is nowhere zero. g F

Let us show first that $F \neq \emptyset$. Since we can express every value $g(\vec{e}) \in \mathbb{Z}_k$ as $\bar{i}$ with $|i| < k$ and then put $f(\vec{e}) := i$, there is clearly a map $f\colon \vec{E} \to \mathbb{Z}$ such that $|f(\vec{e})| < k$ for all $\vec{e} \in \vec{E}$ and $\sigma_k \circ f = g$. For each edge $e \in E$, let us choose one of its two orientations and denote this by $\vec{e}$. We may then define $f'\colon \vec{E} \to \mathbb{Z}$ by setting $f'(\vec{e}) := f(\vec{e})$ and $f'(\overleftarrow{e}) := -f(\vec{e})$ for every $e \in E$. Then f' is a function satisfying (F1) and with values in the desired range; it remains to show that $\sigma_k \circ f'$ and g agree not only on the chosen orientations $\vec{e}$ but also on their inverses $\overleftarrow{e}$. Since σ_k is a homomorphism, this is indeed so:

$$(\sigma_k \circ f')(\overleftarrow{e}) = \sigma_k(-f(\vec{e})) = -(\sigma_k \circ f)(\vec{e}) = -g(\vec{e}) = g(\overleftarrow{e})\,.$$

Hence $f' \in F$, so F is indeed non-empty.

Our aim is to find an $f \in F$ that satisfies Kirchhoff's law (F2), and is thus a k-flow. As a candidate, let us consider an $f \in F$ for which the sum f

$$K(f) := \sum_{x \in V} |f(x, V)|$$ K

of all deviations from Kirchhoff's law is least possible. We shall prove that $K(f) = 0$; then, clearly, $f(x, V) = 0$ for every x, as desired.

Suppose $K(f) \neq 0$. Since f satisfies (F1), and hence $\sum_{x \in V} f(x, V) = f(V, V) = 0$, there exists a vertex x with x

$$f(x, V) > 0\,. \tag{1}$$

Let $X \subseteq V$ be the set of all vertices x' for which G contains a walk $x_0 e_0 \dots e_{\ell-1} x_\ell$ from x to x' such that $f(e_i, x_i, x_{i+1}) > 0$ for all $i < \ell$; furthermore, let $X' := X \smallsetminus \{x\}$. X X'

We first show that X' contains a vertex x' with $f(x', V) < 0$. By definition of X, we have $f(e, x', y) \leqslant 0$ for all edges $e = x'y$ such that $x' \in X$ and $y \in \overline{X}$. In particular, this holds for $x' = x$. Thus, (1) implies $f(x, X') > 0$. Then $f(X', x) < 0$ by (F1), as well as $f(X', X') = 0$. Therefore

$$\sum_{x' \in X'} f(x', V) = f(X', V) = f(X', \overline{X}) + f(X', x) + f(X', X') < 0\,,$$

so some $x' \in X'$ must indeed satisfy x'

$$f(x', V) < 0\,. \tag{2}$$

W As $x' \in X$, there is an x–x' walk $W = x_0 e_0 \dots e_{\ell-1} x_\ell$ such that $f(e_i, x_i, x_{i+1}) > 0$ for all $i < \ell$. We now modify f by sending some flow
f' back along W, letting $f' \colon \vec{E} \to \mathbb{Z}$ be given by

$$f' \colon \vec{e} \mapsto \begin{cases} f(\vec{e}) - k & \text{for } \vec{e} = (e_i, x_i, x_{i+1}),\ \ i = 0, \dots, \ell - 1; \\ f(\vec{e}) + k & \text{for } \vec{e} = (e_i, x_{i+1}, x_i),\ \ i = 0, \dots, \ell - 1; \\ f(\vec{e}) & \text{for } e \notin W. \end{cases}$$

By definition of W, we have $|f'(\vec{e})| < k$ for all $\vec{e} \in \vec{E}$. Hence f', like f, lies in F.

How does the modification of f affect K? At all inner vertices v of W, as well as outside W, the deviation from Kirchhoff's law remains unchanged:

$$f'(v, V) = f(v, V) \qquad \text{for all } v \in V \setminus \{x, x'\}. \tag{3}$$

For x and x', on the other hand, we have

$$f'(x, V) = f(x, V) - k \quad \text{and} \quad f'(x', V) = f(x', V) + k\,. \tag{4}$$

Since g is a $\mathbb{Z}_k$-flow and hence

$$\sigma_k(f(x, V)) = g(x, V) = \overline{0} \in \mathbb{Z}_k$$

and

$$\sigma_k(f(x', V)) = g(x', V) = \overline{0} \in \mathbb{Z}_k\,,$$

$f(x, V)$ and $f(x', V)$ are both multiples of k. Thus $f(x, V) \geqslant k$ and $f(x', V) \leqslant -k$, by (1) and (2). But then (4) implies that

$$|f'(x, V)| < |f(x, V)| \quad \text{and} \quad |f'(x', V)| < |f(x', V)|\,.$$

Together with (3), this gives $K(f') < K(f)$, a contradiction to the choice of f.

Therefore $K(f) = 0$ as claimed, and f is indeed a k-flow. □

Since the sum of two circulations with values in $\mathbb{Z}_k$ is another such circulation, $\mathbb{Z}_k$-flows are often easier to construct (by summing over suitable partial flows) than k-flows. In this way, Theorem 6.3.3 may be of considerable help in determining whether or not some given graph has a k-flow. In the following sections we shall meet a number of examples for this.

Although Theorem 6.3.3 tells us whether a given multigraph admits a k-flow (assuming we know the value of its flow-polynomial for $k - 1$), it does not say anything about the number of such flows. This number is also a polynomial in k, whose values can be bounded above and below by the corresponding values of the flow polynomial. See the notes.

6.4 k-Flows for small k

Trivially, a graph has a 1-flow (the empty set) if and only if it has no edges. In this section we collect a few simple examples of sufficient conditions under which a graph has a 2-, 3- or 4-flow. More examples can be found in the exercises.

Proposition 6.4.1. *A graph has a 2-flow if and only if all its degrees are even.*

Proof. By Theorem 6.3.3, a graph $G = (V, E)$ has a 2-flow if and only if it has a $\mathbb{Z}_2$-flow, i.e. if and only if the constant map $\vec{E} \to \mathbb{Z}_2$ with value $\overline{1}$ satisfies (F2). This is the case if and only if all degrees are even. $\square$ (6.3.3)

For the remainder of this chapter, let us call a graph *even* if all its vertex degrees are even. *even graph*

Proposition 6.4.2. *A cubic graph has a 3-flow if and only if it is bipartite.*

Proof. Let $G = (V, E)$ be a cubic graph. Let us assume first that G has a 3-flow, and hence also a $\mathbb{Z}_3$-flow f. We show that any cycle $C = x_0 \dots x_\ell x_0$ in G has even length (cf. Proposition 1.6.1). Consider two consecutive edges on C, say $e_{i-1} := x_{i-1}x_i$ and $e_i := x_i x_{i+1}$. If f assigned the same value to these edges in the direction of the forward orientation of C, i.e. if $f(e_{i-1}, x_{i-1}, x_i) = f(e_i, x_i, x_{i+1})$, then f could not satisfy (F2) at x_i for any non-zero value of the third edge at x_i. Therefore f assigns the values $\overline{1}$ and $\overline{2}$ to the edges of C alternately, and in particular C has even length. (1.6.1) (6.3.3)

Conversely, let G be bipartite, with vertex bipartition $\{X, Y\}$. Since G is cubic, the map $\vec{E} \to \mathbb{Z}_3$ defined by $f(e, x, y) := \overline{1}$ and $f(e, y, x) := \overline{2}$ for all edges $e = xy$ with $x \in X$ and $y \in Y$ is a $\mathbb{Z}_3$-flow on G. By Theorem 6.3.3, then, G has a 3-flow. $\square$

What are the flow numbers of the complete graphs K^n? For odd $n > 1$, we have $\varphi(K^n) = 2$ by Proposition 6.4.1. Moreover, $\varphi(K^2) = \infty$, and $\varphi(K^4) = 4$; this is easy to see directly (and it follows from Propositions 6.4.2 and 6.4.5). Interestingly, K^4 is the only complete graph with flow number 4:

Proposition 6.4.3. *For all even $n > 4$, $\varphi(K^n) = 3$.*

Proof. Proposition 6.4.1 implies that $\varphi(K^n) \geqslant 3$ for even n. We show, by induction on n, that every $G = K^n$ with even $n > 4$ has a 3-flow. (6.3.3)